AnchorScore: Measuring Faithfulness in Document Question Answering

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Abstract

This paper introduces AnchorScore, a simple faithfulness metric for systems that answer questions from an uploaded document. We built a test set of 200 questions written over 12 computer-science papers. When the system was required to answer from retrieved passages, faithfulness was 91 percent. When the same questions were answered with no document context, faithfulness fell to 54 percent. A chunk size of 800 tokens gave the best recall. Cutting chunks to 200 tokens lowered recall by 11 percentage points. Thirty-eight graduate students rated answer helpfulness at a mean of 4.2 out of 5. We recommend reporting AnchorScore beside accuracy whenever a system answers from uploaded files.

1. Introduction

Document assistants often sound confident even when the answer is not supported by the file the user uploaded. Accuracy against a gold label does not reveal that failure. AnchorScore checks whether each factual claim in an answer appears in the retrieved passages. A claim is anchored only when a human rater can point to a sentence in those passages that states the same fact. The score is the percentage of claims that are anchored.

The study was run at Harborview Research Lab between March and June 2025. The only language in the corpus is English. The only domain is computer science. Those two limits are stated again in Section 5 so they are easy to find.

2. Test collection

Twelve papers were chosen from an internal reading list. None of them is a famous benchmark paper, so memorized model knowledge is a weak substitute for the file. Two annotators wrote 200 questions. Each question has one short factual answer and a pointer to the supporting sentence. Disagreements were resolved by a third annotator. The question types are definition (40), numeric result (70), method detail (50), and limitation (40).

The numeric subset is the hardest slice. It includes counts, percentages, and named thresholds such as the 800-token chunk size. A system that rounds 91 percent to about 90 percent is marked unanchored on that claim.

3. Method

Passages are embedded with all-MiniLM-L6-v2, which produces 384-dimensional vectors. The index is a FAISS flat L2 index. At question time the retriever returns the top 4 passages. The generator may use only those passages. Overlap between adjacent chunks is 150 characters when the chunk size is 1000 characters. That overlap setting is separate from the 800-token experiment in Section 4.

AnchorScore is computed after generation. Raters split the answer into claims, then mark each claim anchored or unsupported. The metric ignores tone, grammar, and answer length. A correct fact that is not present in the retrieved passages still counts as unsupported, because the assistant was not allowed to leave the document.

4. Experiments

Condition A used retrieval. Condition B used the same generator with an empty document. Condition A reached 91 percent faithfulness. Condition B reached 54 percent faithfulness. The 37-point gap is the main result of the paper. On the numeric subset alone, Condition A scored 88 percent and Condition B scored 41 percent.

We also varied chunk size while holding the retriever fixed. Chunks of 200, 400, 800, and 1200 tokens were compared on recall of the supporting sentence. The 800-token setting was best. The 200-token setting reduced recall by 11 percentage points relative to 800 tokens. The 1200-token setting was 3 points below 800 because the extra context diluted the match.

Helpfulness was rated by 38 graduate students on a 1-to-5 scale after they read the answer and the source passage. The mean rating was 4.2. The median rating was 4. No rater was an author of this paper.

5. Limitations

The corpus is English only. The corpus is computer science only. AnchorScore needs human raters, so it is not an automatic training loss. The generator was not compared across model vendors. We did not test scanned image PDFs, because the loader expects selectable text.

6. Conclusion

Retrieval raised faithfulness from 54 percent to 91 percent on 200 questions. The best chunk size in this study was 800 tokens. AnchorScore should be reported alongside accuracy whenever a system answers from uploaded documents.

References

[1] Alvarez, S. and Watanabe, K. AnchorScore: Measuring Faithfulness in Document Question Answering. WDI 2025.